

A Study of the Categories used in 'Papers with Code'

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NSLP 2025



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Motivation

Research Software (RS)

- First-class research output, essential across disciplines
- Enhances simulations, data analysis, modeling, visualization, etc.



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Many taxonomies exist for classification

- E.g.: EDAM, CSO, SWO, Paper with Code, etc.

Papers with Code

- Popular open resource linking ML papers with implementations
- Facilitates exploration of ML methods and datasets
- Features a crowdsourced taxonomy of over 2,000 ML methods



Papers With
Code

Methods

2,344 machine learning components

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Are the methods appropriate for RS classification?



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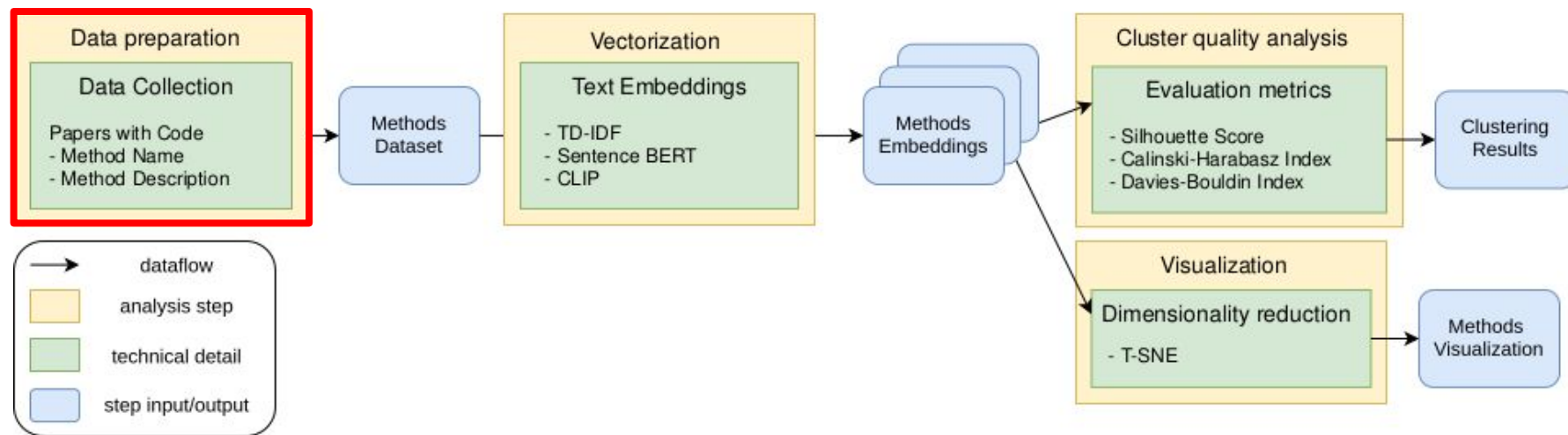
Contribution

- **Methodology** for evaluating the coherence of research software categories
- **Case study:** Papers with Code



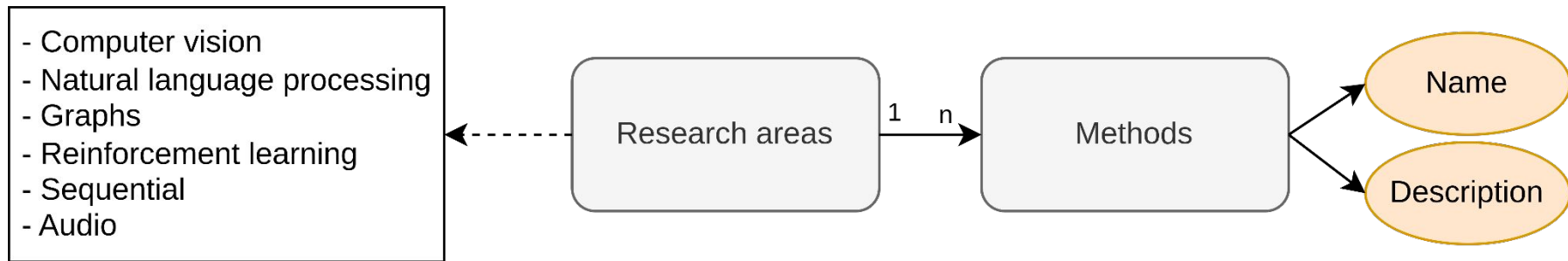
Methodology

Evaluating the coherence of RS categories



Data Preparation

Papers with Code Taxonomy



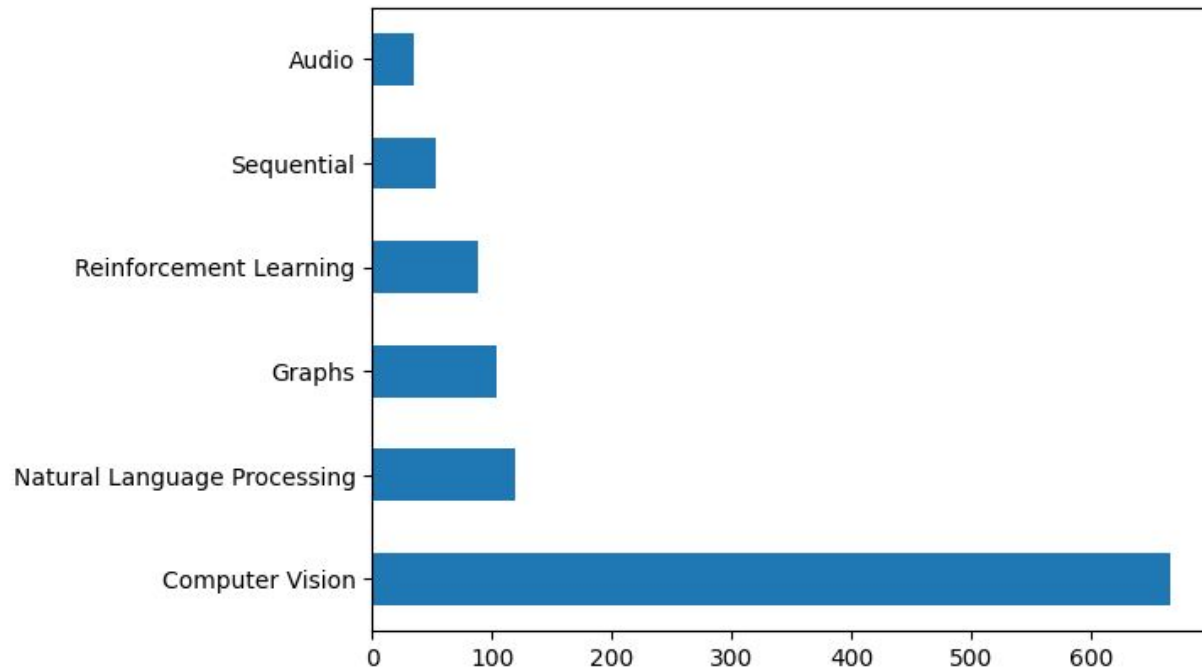
Area: Computer Vision

Name: Inception Module

Description: An **Inception Module** is an image model block that aims to approximate an optimal local sparse structure in a CNN. Put simply, it allows for us to use multiple types of filter size, instead of being restricted to a single filter size, in a single image block, which we then concatenate and pass onto the next layer.

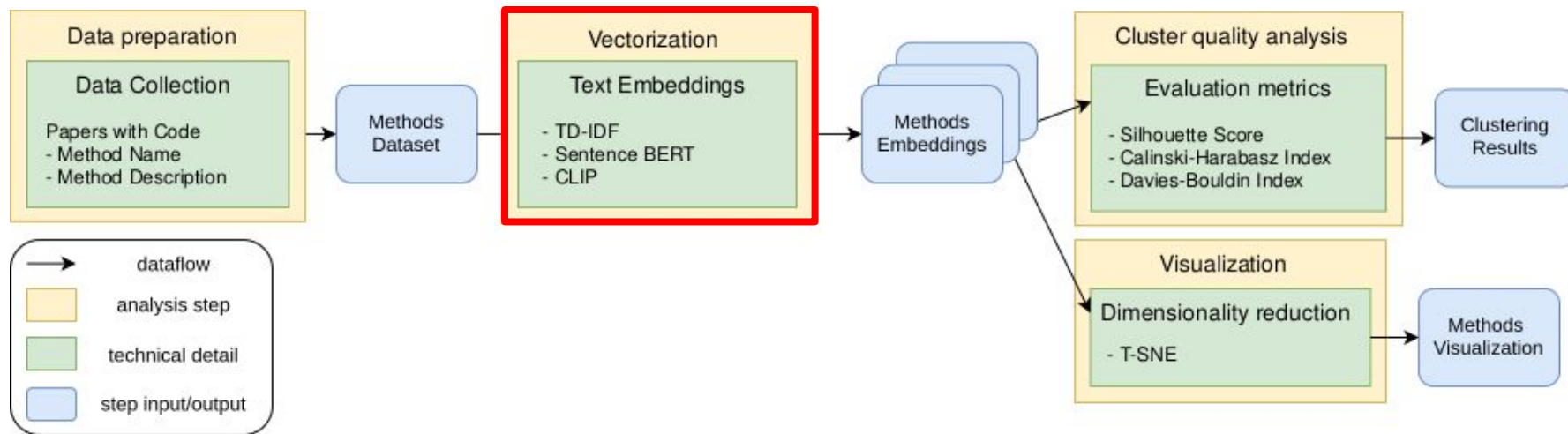
Data Preparation

Research Area - Method distribution



Methodology

Evaluating the coherence of RS categories



Vectorization

Why Use Multiple Methods?

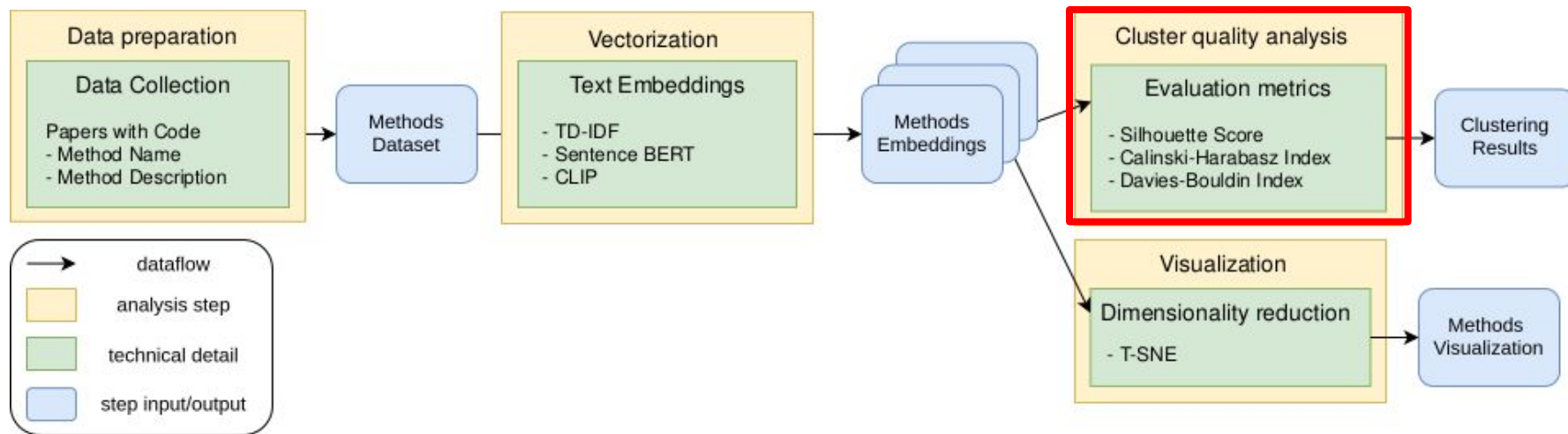
Each method captures different textual features:

- **TF-IDF**: Term frequency - Inverse Document Frequency
 - keyword relevance
- **SBERT**: Sentence Bidirectional Encoder Representations from Transformers
 - semantic similarity
- **CLIP**: Contrastive Language - Image Pretraining
 - high-level contextual meaning

Cross-method comparison improves robustness of results

Methodology

Evaluating the coherence of RS categories

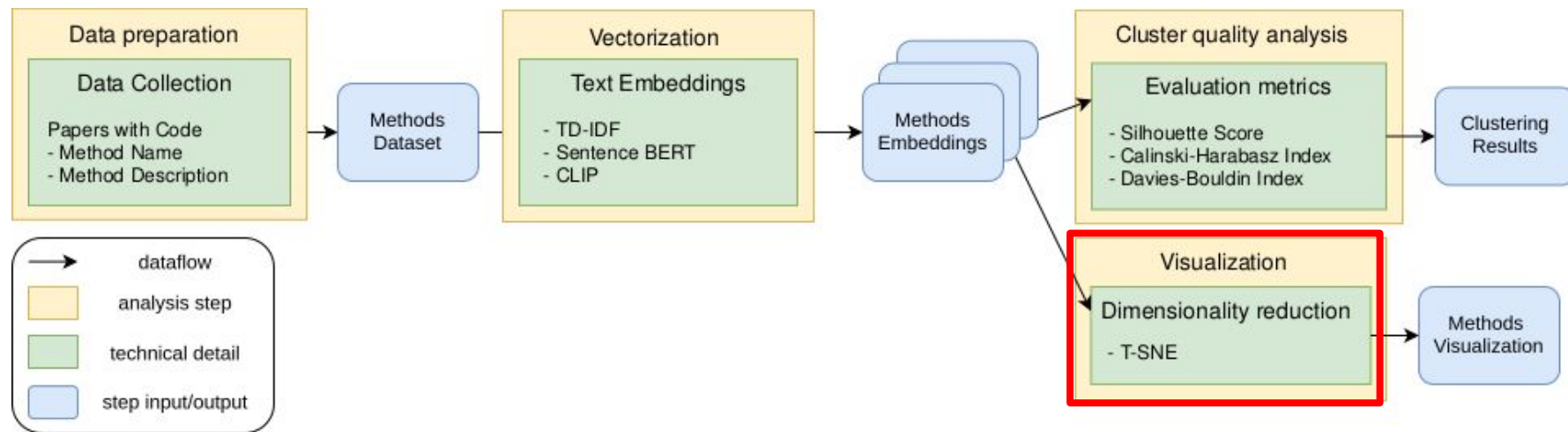


Clustering analysis

- **Silhouette Score (SS)** $[-1, 1]$
 - Measures how well each point fits in its cluster
 - Higher values = better separation
- **Calinski-Harabasz Index (CHI)** $[0, +\infty)$
 - Assesses cluster compactness and separation
 - Higher values = denser, more distinct clusters
- **Davies-Bouldin Index (DBI)** $[0, +\infty)$
 - Measures similarity between clusters
 - Lower values = better separation

Methodology

Evaluating the coherence of RS categories



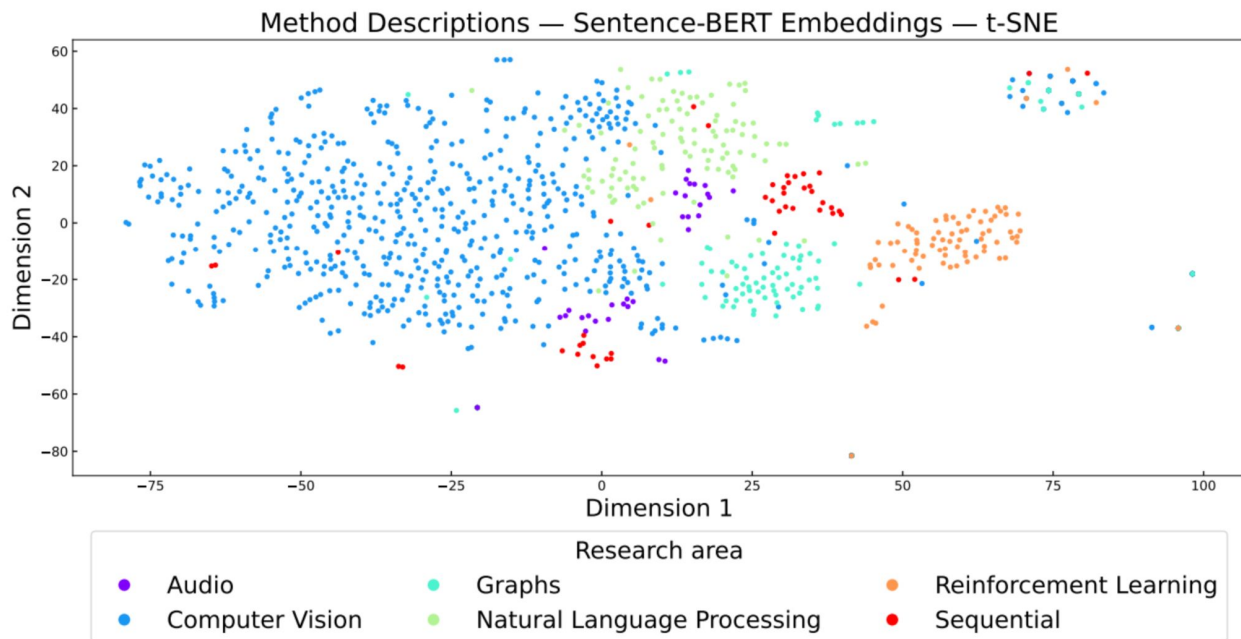
Results

- **Sentence-BERT** performed best across all metrics:
 - Highest cluster quality (CHI)
 - Best separation and compactness (DBI)
- **TF-IDF** and **CLIP** showed weaker separation:
 - Lower CHI, higher DBI, and poor silhouette scores

Embedding	SS		CHI		DBI	
	Name	Desc.	Name	Desc.	Name	Desc.
TF-IDF	-0.10	-0.13	2.20	18.93	25.11	9.40
S-BERT	-0.11	-0.01	6.96	67.37	21.54	3.56
CLIP	-0.10	-0.08	6.31	19.18	10.66	6.15

Results

- Some categories like **Reinforcement Learning** and **NLP** form distinct groups
- Others, like **Computer vision** and **Sequential**, show significant overlap



Results - Summary

- Overlapping terminology reduces category clarity
- Indicates limitations in the current taxonomy
- Suggests the need for refined subcategories and names/descriptions

Conclusions & Future work

- Community-defined categories (e.g., in *Papers with Code*) provide a **useful starting point** for research software classification
- **Clustering analysis** shows that some categories are well-defined
- Others show **significant overlap**
- Low silhouette scores across all embeddings suggest **fuzzy category boundaries**
- The current taxonomy may introduce noise and ambiguity in downstream tasks

Conclusions & Future work

- **Refine category definitions** by:
 - Adding subcategories
 - Clarifying descriptions
 - Including representative examples
- Explore **hierarchical or multi-label classification** approaches to handle cross-disciplinary methods
- Align refined categories with **existing knowledge graphs** to enhance interoperability and findability

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